Homework 10 - Biostatistics

Dylan Renard

Professor Proschan & Professor Nason

JHU Biostatistics

July 21st, 2025

12.

In a study of intra-observer variability in the assessment of cervical smears, 3325 slides were screened for the presence or absence of abnormal squamous cells. Each slide was screened by a particular observer and then rescreened six months later by the same observer. The results of this study are shown below:

First Screening	2nd: present	2nd: absent	Total
Present	1763	489	2252
Absent	403	670	1073
Total	2166	1159	3325

Last week I asked you if these data support the null hypothesis that here is no association between time of screening and diagnosis. This would be a test of independence, such as a chi-square test.

This week we learned about the McNemar's Test for paired data. McNemar's tests a somewhat different null hypothesis: that the true proportion of slides that observer 1 would call positive is the same as the true proportion of slides that observer 2 would call positive. Use McNemar's to see if there is a significant difference in the proportion that the two observers call positive.

McNemar's Test for Paired Data

We are analyzing paired binary data where the same slides were assessed twice by the same observer. Our null hypothesis is:

$$H_0 : p_b = p_c$$

where

- p_b = probability a slide is classified as *present* in the first screening but *absent* in the second,
- p_c = probability a slide is classified as *absent* in the first screening but *present* in the second.

The McNemar test focuses only on these **discordant pairs**, because the slides that were classified the same way both times (the diagonal cells of the 2×2 table) do not provide evidence about disagreement.

From the data:

	2nd: present	**2nd: absent**
1st: present	1763	**489**
1st: absent	**403**	670

We have: - $b = 489$ (present → absent), - $c = 403$ (absent → present).

Test Statistic

For large samples, the McNemar statistic with continuity correction is:

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c}$$

Substituting (b = 489) and (c = 403):

$$\chi^2 = \frac{(|489 - 403| - 1)^2}{489 + 403} = \frac{(86 - 1)^2}{892} = \frac{85^2}{892} = 8.10$$

This follows a chi-square distribution with 1 degree of freedom.

P-value

$$p = 1 - P(\chi^2_{(1)} \leq 8.10) \approx 0.0044$$

Conclusion

Since $p < 0.05$, we reject H_0. There is significant evidence that the proportion of slides classified as positive differs between the first and second screening.

R Code to check work

```r
b <- 489
c <- 403
chi_sq <- (abs(b - c) - 1)^2 / (b + c)
p_value <- 1 - pchisq(chi_sq, df = 1)

chi_sq
```

```
[1] 8.099776
```

```r
p_value
```

```
[1] 0.004427073
```

15.

In an effort to evaluate the health effects of air pollutants containing sulphur, individuals living near a pulp mill in Finland were questioned about various symptoms following a strong emissions released in September 1987. The same subjects were questioned again 4 months later, during a low-exposure period. A summary of the responses related to the occurrence of headaches is presented below.

Low Exposure	High Exposure Yes	High Exposure No	Total
Yes	2	2	4
No	8	33	41
Total	10	35	45

a. Test the null hypothesis that there is no association between exposure to air pollutants containing sulphur and the occurrence of headaches.

Since the **same subjects** were questioned during high exposure and again during low exposure, this is **paired binary data**.
To test whether the proportion of people reporting headaches differs between exposure levels, we use **McNemar's Test**.

Data

For McNemar's test, we focus on the **discordant pairs**: - $b = 2$: Headache during **high exposure** but **not** during low exposure. - $c = 8$: Headache during **low exposure** but **not** during high exposure.

Hypotheses

$$H_0 : p_b = p_c$$

The null hypothesis states that the probability of reporting a headache is the same during high exposure as during low exposure.

Test Statistic

For large samples, McNemar's statistic with continuity correction is:

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c}.$$

Substituting $b = 2$ and $c = 8$:

$$\chi^2 = \frac{(|2-8|-1)^2}{2+8} = \frac{(6-1)^2}{10} = \frac{5^2}{10} = 2.5.$$

P-value

With $df = 1$:

$$p = 1 - P(\chi^2_{(1)} \leq 2.5) \approx 0.114.$$

R Code to check work

```r
b <- 2
c <- 8
chi_sq <- (abs(b - c) - 1)^2 / (b + c)
p_value <- 1 - pchisq(chi_sq, df = 1)

chi_sq
```

```
[1] 2.5
```

```r
p_value
```

```
[1] 0.1138463
```

b. What do you conclude?

The McNemar's test yielded $\chi^2 = 2.5$ with $p \approx 0.114$.
Since $p > 0.05$, we **fail to reject** H_0.

Interpretation: There is insufficient evidence to conclude that the probability of reporting headaches differs between the high- and low-exposure periods for the same subjects.

c. If you found a significant association, what limitations of this study would affect your ability to conclude that pollution caused headaches?

Even if we had found a significant association between sulphur pollution exposure and headaches, the study design has several limitations that would affect our ability to conclude that pollution caused the headaches:

1. **Observational Study Design:**
   There was no random assignment of exposure levels. Without randomization, other factors (e.g., stress, seasonal differences, or concurrent illnesses) could confound the results.

2. **Self-Reported Symptoms:**
   Headache data are based on self-reports, which can be subjective and prone to recall bias.

3. **Temporal Changes:**
   The second survey was conducted four months later during a different season, and seasonal

factors (such as weather or allergens) might influence headaches.

4. **No Control Group:**
   The same subjects were measured twice, but there was no group of individuals unexposed to sulphur pollution over the same period to serve as a control.

5. **Small Sample Size:**
   With only 45 subjects, random variation and sampling error are large, reducing the reliability of the observed association.

Conclusion: Even with a statistically significant result, these limitations mean we could not infer a causal relationship between pollution exposure and headaches.

17.

In France, a study was conducted to investigate potential risk factors for ectopic pregnancy. Of the 279 women who had experienced ectopic pregnancy, 28 had suffered from pelvic inflammatory disease. Of the 279 women who had not, 6 had suffered from pelvic inflammatory disease.

a. Construct a 2x2 contingency table

```
# Load the DT package
library(DT)

# Data values
a <- 28   # Ectopic & PID
b <- 251  # Ectopic & no PID
c <- 6    # No Ectopic & PID
d <- 273  # No Ectopic & no PID

# Create the 2x2 contingency table
contingency_table <- matrix(c(a, b, c, d), nrow = 2, byrow = TRUE)

# Add row and column names
rownames(contingency_table) <- c("Ectopic Pregnancy", "No Ectopic Pregnancy")
colnames(contingency_table) <- c("PID (Yes)", "PID (No)")

# Convert to data frame for better formatting
contingency_df <- as.data.frame(contingency_table)
contingency_df <- cbind(Row = rownames(contingency_table), contingency_df)

# Display interactive table
datatable(
  contingency_df,
  rownames = FALSE,
  options = list(
    dom = 't',
    paging = FALSE,
    ordering = FALSE
```

```
  )
)
```

Row	PID (Yes)	PID (No)
Ectopic Pregnancy	28	251
No Ectopic Pregnancy	6	273

b. Estimate the odds ratio of suffering ectopic pregnancy for women who had pelvic inflammatory disease vs women who have not.

The odds ratio (OR) for ectopic pregnancy comparing women with pelvic inflammatory disease (PID) to those without PID is calculated as:

$$\text{OR} = \frac{a \cdot d}{b \cdot c},$$

where:

- $a = 28$ (Ectopic & PID),
- $b = 251$ (Ectopic & No PID),
- $c = 6$ (No Ectopic & PID),
- $d = 273$ (No Ectopic & No PID).

Substituting the values:

$$\text{OR} = \frac{28 \cdot 273}{251 \cdot 6} = \frac{7644}{1506} \approx 5.07.$$

R Code to check work

```
# Data values
a <- 28
b <- 251
c <- 6
d <- 273

# Odds Ratio
OR <- (a * d) / (b * c)
OR
```

```
[1] 5.075697
```

Interpretation:
Women with pelvic inflammatory disease are estimated to have approximately **5 times the odds** of ectopic pregnancy compared to women without PID.

c. Find a 99% confidence interval for the population odds ratio.

17(c) 99% Confidence Interval for the Population Odds Ratio

To find the confidence interval (CI) for the population odds ratio (OR), we use the log transformation:

1. **Log of the OR:**

$$\ln(\text{OR}) = \ln\left(\frac{a \cdot d}{b \cdot c}\right)$$

2. **Standard Error (SE):**

$$SE = \sqrt{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}}$$

3. **99% CI for log(OR):**

$$\ln(\text{OR}) \pm z_{0.005} \cdot SE$$

where

$$z_{0.005} = 2.576$$

for a 99% confidence interval.

4. **Exponentiate the bounds** to obtain the CI for OR.

Step-by-Step Calculation

Given:

$$a = 28,\ b = 251,\ c = 6,\ d = 273,\ \text{OR} = 5.07.$$

- **Log(OR):**

$$\ln(\text{OR}) = \ln(5.07) \approx 1.622$$

- **Standard Error:**

$$SE = \sqrt{\frac{1}{28} + \frac{1}{251} + \frac{1}{6} + \frac{1}{273}} = \sqrt{0.0357 + 0.0040 + 0.1667 + 0.0037} = \sqrt{0.2101} \approx 0.458$$

- **Confidence Limits (log scale):**

$$1.622 \pm 2.576 \cdot 0.458 = 1.622 \pm 1.18$$

which gives:

$$[0.442,\ 2.802]$$

- **Confidence Limits (OR scale):**

$$[e^{0.442},\, e^{2.802}] \approx [1.55,\, 16.47]$$

Interpretation: We are 99% confident that the population odds ratio lies between **1.55** and **16.47**.

d. Assuming the women were chosen to have equal numbers who had and had not experienced an ectopic pregnancy, what kind of study is this? Is the relative risk that you can calculate from this table meaningful?

Since the study deliberately selected **equal numbers of women with ectopic pregnancy (cases) and women without ectopic pregnancy (controls)**, this is a **case-control study**.

In a case-control study:

- The number of cases and controls is **fixed by design**, not by the natural frequency of the disease in the population.
- The marginal totals of the table (i.e., the row sums) **do not represent population prevalence** or risk.

Why Relative Risk is Not Meaningful Here

The **relative risk (RR)** is defined as:

$$RR = \frac{P(\text{disease} \mid \text{exposed})}{P(\text{disease} \mid \text{not exposed})}.$$

To compute this, we need the **true probabilities** of disease in exposed and unexposed groups, which requires the sample to reflect population proportions.

However, in a case-control design:

- The sampling artificially forces the number of cases (279) and non-cases (279) to be equal.
- This means $P(\text{disease})$ in this sample is not the true population probability.

As a result, **RR calculated from this table would be biased and meaningless** because:

1. The probability of disease is **overrepresented** (50% in the table vs. a much smaller rate in the real population).
2. The exposure distribution among controls might not reflect that in the overall population.

What Can We Use Instead?

The **odds ratio (OR)** remains valid because: - OR is a ratio of odds, which is invariant to the way we sample cases and controls. - When the disease is rare, $OR \approx RR$, so OR provides a good approximation of RR in the population.

In Conclusion:
This is a **case-control study**, and RR cannot be meaningfully calculated from this data. The **OR is the appropriate measure of association**.

20.

A study was conducted to determine whether geographic variations in the use of medical and surgical services could be explained in part by differences in the appropriateness with which physicians use these services. One concern might be that a high rate of inappropriate use of a service is associated with high overall use within a particular region. For the procedure coronary angiography, three geographic areas were studied: a high-use site (site 1), a low use urban site (site 2), and a low use rural site (site 3). Within each geographical region, each use of this procedure was classified as appropriate, equivocal, or inappropriate by a panel of expert physicians. These data are saved in the data set *angio*. Site number is saved under the variable name *site*, and level of appropriateness under *appropro*.

A. At the 0.05 level of significance, test the null hypothesis that there is no association between geographic region and the appropriateness of use of coronary angiography.

```
angio <- read.table("angio.txt", header = TRUE)

# Contingency table
table_angio <- table(angio$site, angio$appropro)
table_angio
```

```
       appropr equiv inappro
 site1     452    63     113
 site2     394    35      85
 site3     434    21      80
```

```
# Chi-square test
chisq.test(table_angio, correct = FALSE)
```

```
    Pearson's Chi-squared test

data:  table_angio
X-squared = 20.056, df = 4, p-value = 0.0004868
```

B. What do you conclude?

The chi-square test of independence between **geographic site** and **appropriateness of coronary angiography** returned:

- Test statistic: (20.056)
- Degrees of freedom: (4)
- p-value: **0.00049**

Since the p-value is much smaller than 0.05, we **reject** H_0.
Overall, There is strong evidence that the appropriateness of coronary angiography **varies significantly across the three geographic regions**.

C. What is your estimate of the odds ratio for having an inappropriate use of Angiography, comparing the high use urban site to the rural site?

```
# Filter only sites 2 and 3
urban_vs_rural <- table_angio[c("site2", "site3"), ]

# Get counts
inappropriate <- urban_vs_rural[, "inappro"]   # corrected spelling
not_inappropriate <- rowSums(urban_vs_rural[, c("appropr", "equiv")])

# Construct 2x2 table
odds_table <- matrix(c(inappropriate["site2"], not_inappropriate["site2"],
                       inappropriate["site3"], not_inappropriate["site3"]),
                     nrow = 2, byrow = TRUE)
colnames(odds_table) <- c("Inappropriate", "Not_Inappropriate")
rownames(odds_table) <- c("Urban (2)", "Rural (3)")
odds_table
```

```
          Inappropriate Not_Inappropriate
Urban (2)            85                429
Rural (3)            80                455
```

```
# Odds ratio
a <- odds_table[1,1]
b <- odds_table[1,2]
c <- odds_table[2,1]
d <- odds_table[2,2]
OR <- (a * d) / (b * c)
OR
```

```
[1] 1.126894
```

Using the data, we obtained:

$$OR \approx 1.1269.$$

Interpretation:
The odds of inappropriate use at the high-use urban site are about **1.13 times** those at the rural site, indicating only a slight difference.

16.7.

In a study investigating various risk factors for heart disease, the relationship between hypertension and coronary artery disease (CAD) was examined for individuals in two different age groups. (see tables below)

35-49 Years of Age

CAD Yes	CAD No	Total	
Hypertension Yes	552	212	764
Hypertension No	941	495	1436
Total	1493	707	2200

Over 65 Years of Age

CAD Yes	CAD No	Total	
Yes	1102	87	1189
No	1018	106	1124
Total	2120	193	2313

A. Within each category of age, are the odds of suffering from coronary artery disease greater or smaller for individuals with hypertension?

```
### 16.7(A) Odds Ratios for Each Age Group

# 35–49 years of age table
age1 <- matrix(c(552, 212,
                 941, 495),
               nrow = 2, byrow = TRUE)
colnames(age1) <- c("CAD Yes", "CAD No")
rownames(age1) <- c("Hypertension Yes", "Hypertension No")
age1
```

```
                CAD Yes CAD No
Hypertension Yes     552    212
Hypertension No      941    495
```

```
# Over 65 years of age table
age2 <- matrix(c(1102, 87,
                 1018, 106),
               nrow = 2, byrow = TRUE)
colnames(age2) <- c("CAD Yes", "CAD No")
rownames(age2) <- c("Hypertension Yes", "Hypertension No")
age2
```

```
                CAD Yes CAD No
Hypertension Yes    1102     87
Hypertension No     1018    106
```

```
# Function to compute odds ratio
odds_ratio <- function(tab) {
  a <- tab[1, 1]
  b <- tab[1, 2]
  c <- tab[2, 1]
  d <- tab[2, 2]
  OR <- (a * d) / (b * c)
  return(OR)
}

# Calculate ORs
OR_age1 <- odds_ratio(age1)
OR_age2 <- odds_ratio(age2)

OR_age1
```

```
[1] 1.369679
```

```
OR_age2
```

```
[1] 1.318926
```

16.7(A) Conclusion

- **Ages 35–49:**
  The odds ratio is approximately **1.36**, indicating that individuals with hypertension have **1.36 times the odds** of having coronary artery disease (CAD) compared to those without hypertension.

- **Ages over 65:**
  The odds ratio is approximately **1.31**, indicating that individuals with hypertension have **1.31 times the odds** of having CAD compared to those without hypertension.

B. Do you feel that it is appropriate to combine the information in these two tables to make a single unifying statement about the relationship between hypertension and coronary artery disease? Why or why not?

Yes, it is reasonable to combine the information from the two tables to make a single unifying statement about the relationship between hypertension and coronary artery disease (CAD).

Why? - The odds ratios for the two age groups are very similar (1.36 for ages 35–49 and 1.31 for ages over 65). - This suggests that the association between hypertension and CAD is consistent (homogeneous) across the age strata. - When the effect size is similar across groups, a stratified analysis (e.g., using the Mantel–Haenszel method) provides a valid summary measure that accounts for age while estimating the overall relationship.

However: If the odds ratios were very different between age groups, it would not be appropriate to combine the tables because the association would depend on age (effect modification).

C. Compute a Mantel -Haenszel estimate of the summary odds ratio.

Using the formula:

$$\hat{\theta}_{MH} = \frac{\sum_i \frac{a_i d_i}{n_i}}{\sum_i \frac{b_i c_i}{n_i}},$$

we compute:

- For ages 35–49:

$$\frac{a_1 d_1}{n_1} \approx 124.2, \quad \frac{b_1 c_1}{n_1} \approx 90.6.$$

- For ages >65:

$$\frac{a_2 d_2}{n_2} \approx 50.5, \quad \frac{b_2 c_2}{n_2} \approx 38.3.$$

Thus:

$$\hat{\theta}_{MH} = \frac{124.2 + 50.5}{90.6 + 38.3} \approx 1.36.$$

R Code to check work:

```r
# Construct the 2x2 tables for each stratum
age1 <- matrix(c(552, 212,
                 941, 495),
               nrow = 2, byrow = TRUE,
               dimnames = list(Hypertension = c("Yes", "No"),
                               CAD = c("Yes", "No")))

age2 <- matrix(c(1102, 87,
                 1018, 106),
               nrow = 2, byrow = TRUE,
               dimnames = list(Hypertension = c("Yes", "No"),
                               CAD = c("Yes", "No")))

# Combine into an array for mantelhaen.test
tables <- array(c(age1, age2), dim = c(2, 2, 2),
                dimnames = list(
                  Hypertension = c("Yes", "No"),
                  CAD = c("Yes", "No"),
                  AgeGroup = c("35-49", "Over65")
                ))

# Mantel-Haenszel test
mh_test <- mantelhaen.test(tables)
mh_test
```

```
    Mantel-Haenszel chi-squared test with continuity correction
```

```
data:  tables
Mantel-Haenszel X-squared = 13.372, df = 1, p-value = 0.0002554
alternative hypothesis: true common odds ratio is not equal to 1
95 percent confidence interval:
 1.152997 1.591479
sample estimates:
common odds ratio
         1.354611
```

Conclusion:
The Mantel–Haenszel estimate of the summary odds ratio is approximately **1.36**.

D. Construct a 95% confidence interval for the summary odds ratio.

From `mantelhaen.test`, the Mantel–Haenszel summary odds ratio is:

$$\hat{\theta}_{MH} = 1.36,$$

with a 95% confidence interval of:

$$[1.153, 1.591].$$

Conclusion: We are 95% confident that the odds of CAD are between **1.15 and 1.59** times higher for individuals with hypertension, after adjusting for age.

E. At the alpha= .05 level of significance, Test the null hypothesis that there is no association between hypertension and coronary artery disease. What do you conclude?

Null hypothesis H_0:
There is no association between hypertension and coronary artery disease (CAD), i.e., the common odds ratio $\theta_{MH} = 1$.

Alternative hypothesis H_A:
There is an association between hypertension and CAD ($\theta_{MH} \neq 1$)).

Mantel–Haenszel Test

Using the `mantelhaen.test` function in R:

```
mh_test <- mantelhaen.test(tables)
mh_test
```

```
    Mantel-Haenszel chi-squared test with continuity correction

data:  tables
Mantel-Haenszel X-squared = 13.372, df = 1, p-value = 0.0002554
alternative hypothesis: true common odds ratio is not equal to 1
95 percent confidence interval:
 1.152997 1.591479
```

```
sample estimates:
common odds ratio
         1.354611
```

Conclusion for 16.7(E)

The Mantel–Haenszel test shows a summary odds ratio of approximately **1.36** with a 95% confidence interval of **[1.153, 1.591]** and a p-value less than **0.05**.
Since $p < 0.05$, we **reject** H_0 and conclude that there is a statistically significant association between hypertension and coronary artery disease, after adjusting for age.